

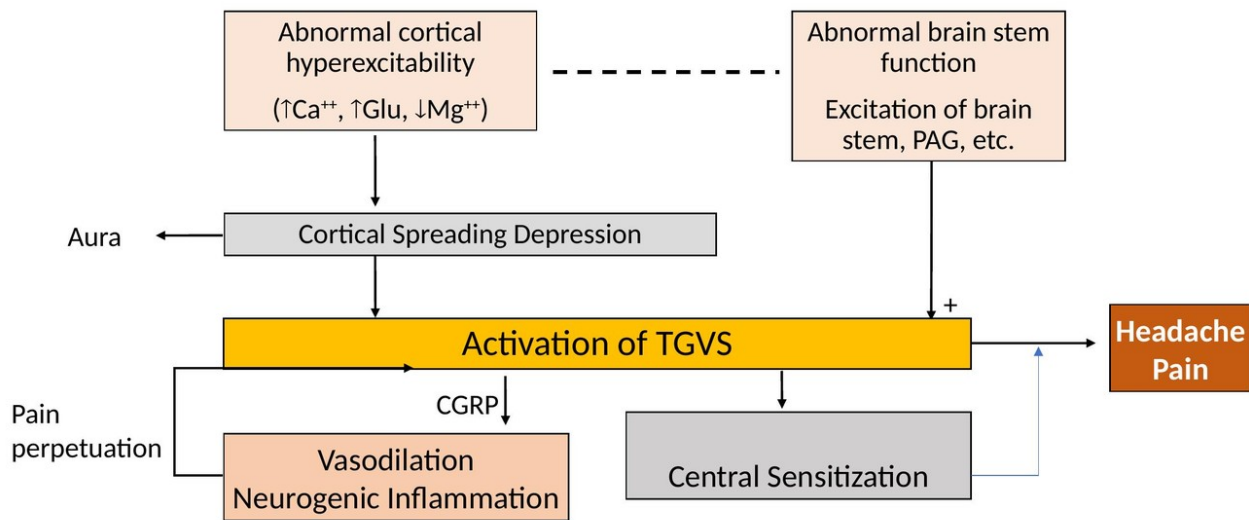
Lecture 3 — Headache

Companion Figures

Figures referenced from the Notion revision page.



Migraine: Central Hypothesis Pathophysiology



PAG = periaqueductal gray, TGVS = trigemino-vascular system. CGRP = Calcitonin gene-related peptide

Adapted from Pietrobon D, Striessnig J. *Nat Rev Neurosci.* 2003;4:386-398.

Organisation of the nervous system: Compare and contrast the structure of the central, peripheral and autonomic (sympathetic and parasympathetic) nervous systems.

Nervous system disorders: Summarise the aetiology and pathophysiology of central and peripheral nervous system disorders.

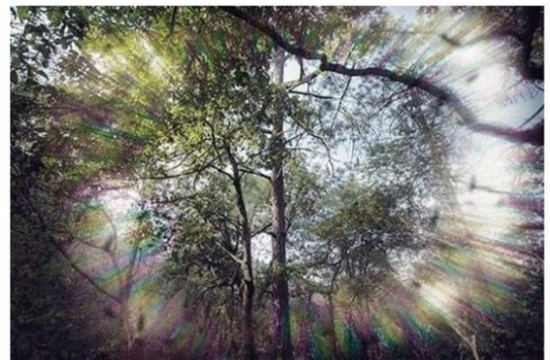
Figure L3.8.1 — Migraine central-hypothesis pathophysiology

Cortical hyperexcitability + brainstem trigger → cortical spreading depression → aura (branch) and TGVS activation → CGRP → vasodilation + neurogenic inflammation + central sensitisation → headache pain.

selves through the usual round of work and play, a degree of
ness and a desire for rest are characteristic of severe
migraine. A vascular head is exquisitely sensitive to motion
head may in itself enforce activity, but we find it not
only, or even the chief, mechanism at work. Many patients
during an attack and exhibit diminished tone of skeletal
Many are dejected, and seek seclusion and passivity. A
drowsy.

The relation of sleep to the complex and func-
one, and we will have to touch upon it in many
contexts: the in- scope and stupor in the acute
migraine (migr- and classical migraine), the tend
migraines of occur during sleep, and their
relation to these states. At this point we r
attention to the relationship: the oc
of intense dro a common r
the occasional abo sleep of unusual
and the typical protracted up in which many attacks fi
natural termination.

Nowhere in the literature can we find more vivid and
descriptions of migrainous stupor than in Liveing's monogr



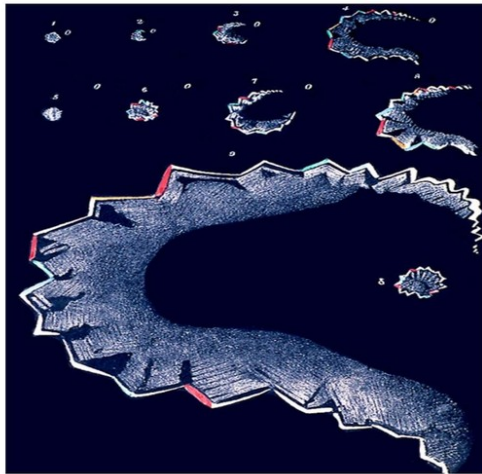
pheral and autonomic (sympathetic and parasympathetic) nervous systems.
ripheral nervous system disorders.

Figure L3.8.2 — Visual aura — historical description and modern reconstruction

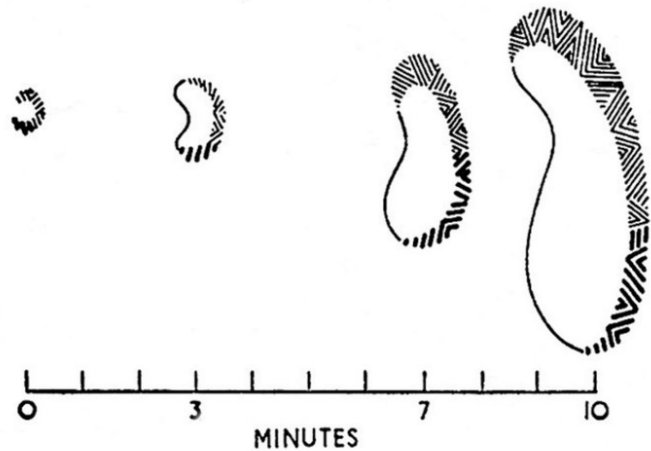
Liveing's 1873 text partially obscured by the shape of a scintillating scotoma, with a modern photograph showing how an expanding shimmering arc distorts the visual field during aura.

Migraine: aura

Expanding 'C's



Elemental visual disturbance

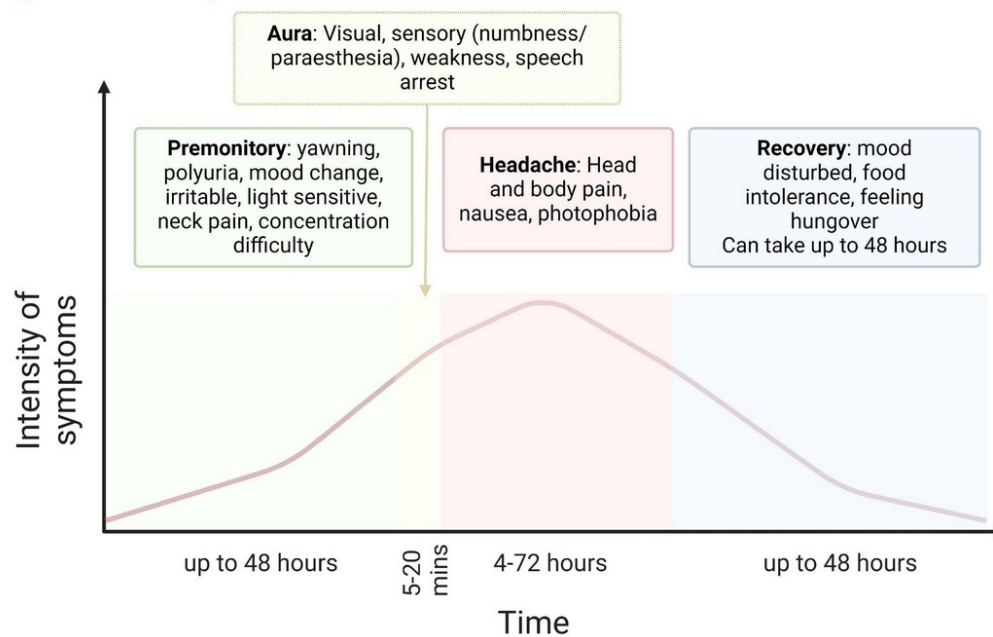


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Figure L3.8.3 — Two morphologies of visual aura

Left: expanding "C"-shaped fortification spectra. Right: elemental visual disturbance enlarging gradually over ~10 minutes. Both are CSD crossing the visual cortex, both evolve slowly — sudden onset is NOT aura.

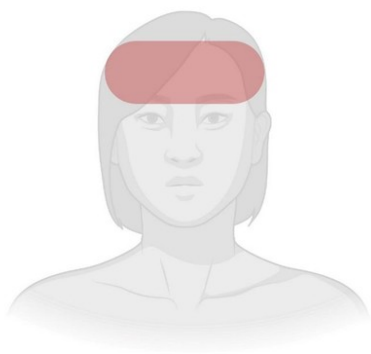
Migraine: phases



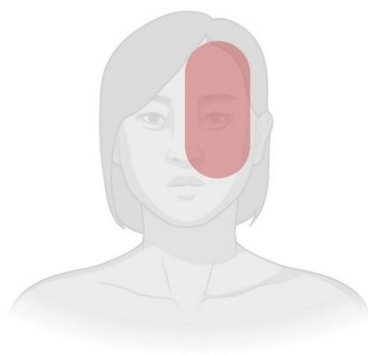
ation of the nervous system: Compare and contrast the structure of the central, peripheral and autonomic (sympathetic and parasympathetic) nervous systems.
is system disorders: Summarise the pathology and pathophysiology of central and peripheral nervous system disorders.

Figure L3.8.4 — Migraine phases — symptom intensity over time

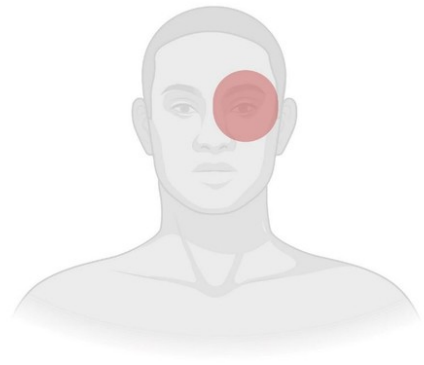
Four phases: premonitory (up to 48 h) → aura (5–20 min, if present) → headache (4–72 h) → recovery (up to 48 h). The full "migraine journey" can span a week.



Tension



Migraine



Cluster

Figure L3.8.5 — Pain distribution in the three primary headaches

Tension — bilateral band across the forehead. Migraine — unilateral, broad hemiface. Cluster — unilateral, focused around one eye (V1).